

ALDO VILLAMIL

AI Architek Engineer · Full-Stack Engineer · Web3 Builder · Co-Founder

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SUMMARY

AI Architek Engineer — senior systems builder, bilingual (EN/ES), 20+ yrs professional background. I design and run autonomous multi-agent systems in production. I architect an **orchestrating harness** that routes work to specialist agents (OpenCode, Cline, Codex) as a coordinated swarm, backs it with a **council of specialist models** — each LLM treated as an independent policy, aggregated by a reward model into a single high-confidence answer — and keeps design intent alive across sessions with **memory engineered through an automated write path** (facts, not logs). A fail-closed **security lane (Sentinel)** watches network exposure, secrets, credentials, and spend 24/7. I ship full products end-to-end — **CRM & client portals, data-rich dashboards, enterprise ecosystems, and on-chain payment rails** where autonomous agents pay each other (x402) — deployed on Vercel and DigitalOcean. Fluent in English and Spanish, US-timezone experienced, comfortable owning architecture through deployment.

AI ARCHITECTURE & AGENT SYSTEMS

Autonomous multi-agent swarm (production)

- An orchestrating harness decides, routes, and verifies; specialist sub-agents (OpenCode, Cline, Codex) build in parallel — one task per agent, isolated workspaces, no self-report trusted until verified.
- Runs 24/7 on production cloud infrastructure over an encrypted private network; results supervised and verified end-to-end.

Council of models (reliability at the answer level)

- For high-stakes decisions, the same question is routed to a council of specialist LLMs, outputs aggregated with a reward model, and the single highest-confidence answer returned — no single model trusted blindly.

Memory engineering (council + automated write path)

- Cross-session memory keeps requirements, decisions, and design intent intact across every build.
- An automated seed write-path distills source-of-truth docs into dated facts and skills — engineered to forget (dedup, consolidation, revocation) so the system stays decision-relevant, never a growing bucket.

Verification-first pipeline

- Spec-first workflow: capture intent, orchestrate specialist agents, verify every artifact against a 3-stage gate (lint → typecheck → integration) before it ships, then deploy and self-monitor.

Security posture — Sentinel (always-on, fail-closed)

- A dedicated security lane monitors network exposure, secret-scanning across repos and logs, credential-file permissions, and spend.
- Fail-closed by design: any anomaly escalates; secrets live in env only and are redacted from all output; memory is treated as untrusted input until validated.

Isolated agent workspaces

- Fully isolated environments — own memory, sessions, tooling, secrets — zero cross-contamination across projects, coordinated as one swarm.

TECHNICAL STACK

AI / AGENTS: DeepSeek V4 · OpenRouter · Groq · Hermes Agent · OpenCode · Cline · Codex · multi-model consensus · MCP · A2A · DeepCode

ML / RL: PPO · GRPO · RLVR · reward-model consensus · fine-tuning

LANGUAGES: TypeScript · Python · Solidity · SQL · Rust

FRONTEND: Next.js · React · Tailwind CSS · Three.js / R3F

BACKEND / DB: Prisma · Neon PostgreSQL · Redis (Upstash) · Supabase

AUTOMATION: n8n · DeerFlow · VoltAgent · Cron Jobs · Message Queues

DEVOPS: GitHub Actions CI/CD · Vercel Cron · Sentry · Health Checks · Upstash rate limiting · Sentinel

PAYMENTS: Wompi · Stripe · x402 · c402 (HTTP 402) · PSE / Nequi

BLOCKCHAIN: Base · Ethereum · Solana · Chainlink VRF · ERC-1155

CLOUD / INFRA: Vercel · DigitalOcean · Docker · Kubernetes (AMD Instinct) · Nginx

IDENTITY: ENS · Farcaster · Clerk · IPFS

FLAGSHIP PROJECTS

MCPVOTS — on-chain x402 facilitator · Solidity · Base · Chainlink VRF · ERC-1155 · USDC

Official x402 facilitator enabling autonomous agent-to-agent payments on Base. ERC-1155 with Chainlink VRF for provably fair draws, settled in USDC — "Stripe for AI agents." Coinbase-ecosystem listed, full Hardhat test coverage.

CRM & CLIENT PLATFORMS — Next.js · React · Rust · PostgreSQL · Redis

Production CRM and client-facing portals: multi-tenant dashboards, real-time metrics, role-based access, payment rails, and agent-assisted workflows — delivered with an internal team.

MCPVOTSAGI — multi-model consensus SDK · Python · AsyncIO · WebSockets · Redis

Multi-model consensus SDK: specialist LLM outputs aggregated by a reward model into a single high-confidence answer. A2A protocol gateway for agent-to-agent WebSocket communication, Redis-backed persistent memory.

DATA PLATFORMS & APIS — TypeScript · Python · c402 (HTTP 402)

Public data APIs with c402 monetization, live dashboards over rich datasets, and typed SDKs — built for broad industry use.

WORK EXPERIENCE

Co-Founder & Full-Stack Engineer — Village of Thousands 2021 – Present · Remote

- Built headless e-commerce: React + Next.js storefront on Vercel, backend on DigitalOcean (99.9% uptime).
- Created all brand art and visual identity — graffiti-inspired design language; NYFW-featured.

Digital Marketing Manager — PolloRiko.io 2018 – Present · Houston, TX (Remote)

- Own full digital presence for a US restaurant chain: website, social, SEO, AI-generated content.

Customer Service — United Health Care (Teleperformance) 2020 – 2021 · Colombia

- English-language support for US-based members. High CSAT, US time zones.

Banquet Captain — Hilton Bonnet Creek 2017 – 2018 · Orlando, FL

- Led service teams for large-scale corporate events; luxury service standards.

EDUCATION & CERTIFICATIONS

- Computer Programming and Analysis — Valencia Community College, Orlando, FL (2004–2008)
- High School Diploma — Benjamin Cardozo HS, New York City (2000)
- **AMD AI Academy — Certified:** AI on AMD Instinct GPUs · Hugging Face on AMD · Kubernetes on AMD Instinct

Continuous self-directed learning verified through shipped production systems.